

STREAMING EVENT ANOMALY DETECTOR

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1. PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Title

Streaming Event Anomaly Detector with AI Agent

Project Overview

The Streaming Event Anomaly Detector is an AI-powered monitoring platform designed to process streaming events in real time, identify abnormal patterns using statistical anomaly detection techniques, provide AI-generated explanations, and trigger automated alerts for critical situations.

The system enables organizations to monitor operational events continuously and respond proactively to unusual system behavior.

Problem Statement

Modern applications generate thousands of events every second. Monitoring these events manually is difficult and inefficient. Abnormal activities often go unnoticed until they impact system performance, security, or business operations.

The proposed solution automatically analyzes incoming event streams, detects anomalies, explains their causes using Artificial Intelligence, and alerts relevant stakeholders immediately.

Objectives

- Monitor real-time streaming events.
- Detect anomalies automatically.
- Generate AI-powered explanations.

- Classify anomaly severity.
- Trigger automated alerts.
- Provide centralized dashboard monitoring.
- Maintain historical event records.

Target Users

- System Administrators
- DevOps Engineers
- Security Analysts
- Operations Teams
- Monitoring Engineers

Core Features

Event Generation

- Simulates real-time event streams.

Event Processing

- Processes incoming events continuously.

Anomaly Detection

- Identifies unusual behavior using Z-Score calculations.

AI Explanation Engine

- Provides natural language explanations for detected anomalies.

AI Agent

- Performs intelligent decision-making and recommends corrective actions.

Alert Management

- Sends notifications using Discord webhooks.

Dashboard Monitoring

- Displays metrics, trends, and alerts in real time.

Authentication & Security

- Provides JWT-based authentication and role management.

Success Metrics

- Detection Accuracy > 95%

- Alert Delivery Time < 5 Seconds
- Dashboard Availability > 99%
- Real-Time Processing Capability
- Reduced Manual Monitoring Effort

MVP Completion Criteria

- ✓ Event Generation Functional
- ✓ Event Processing Functional
- ✓ Anomaly Detection Working
- ✓ AI Explanations Generated
- ✓ AI Agent Operational
- ✓ Discord Alerts Functional
- ✓ Dashboard Available
- ✓ Authentication Enabled

2. TECHNICAL REQUIREMENTS DOCUMENT (TRD)

Technology Stack

Frontend

- React
- Vite
- JavaScript
- Recharts

Backend

- FastAPI
- Python

Database

- SQLite

AI Components

- Ollama

- Llama 3.1

Security

- JWT Authentication
- Role-Based Access Control (RBAC)

Alerting

- Discord Webhooks

Containerization

- Docker

Architecture Components

Event Producer

- Generates simulated streaming events.

Event Queue

- Stores events temporarily before processing.

Event Consumer

- Reads and processes events from the queue.

Anomaly Detection Engine

- Calculates Z-Scores and identifies anomalies.

AI Explanation Module

- Uses Llama 3.1 to explain anomalies.

AI Agent

- Determines severity levels and recommended actions.

Alert Service

- Sends notifications to Discord channels.

Dashboard Service

- Displays system analytics and monitoring information.

Environment Variables

JWT_SECRET

DATABASE_URL

OLLAMA_URL

DISCORD_WEBHOOK_URL

Non-Functional Requirements

Performance

- Low latency processing
- Real-time updates

Security

- JWT Authentication
- Password Hashing
- Role-Based Authorization

Reliability

- Continuous event processing
- Persistent storage

Scalability

- Future support for Kafka integration
- Future support for PostgreSQL

Future Improvements

- Apache Kafka Integration
- Machine Learning-Based Detection
- PostgreSQL Migration
- Kubernetes Deployment
- Multi-Tenant Support

3. APPLICATION FLOW & USER JOURNEY

Application Pages

Landing Page

Displays project overview and login options.

Login Page

Authenticates users.

Register Page

Allows new user registration.

Dashboard

Displays system metrics and monitoring information.

Event Monitor

Shows incoming event streams.

Anomaly Viewer

Displays detected anomalies.

AI Analysis Panel

Shows AI-generated explanations and recommendations.

Alert Center

Displays alert history and status.

Navigation Structure

Dashboard → Events → Anomalies → AI Analysis → Alerts → Settings

Authentication Flow

Register



Login



Dashboard

Core User Journey

Login



Generate Events

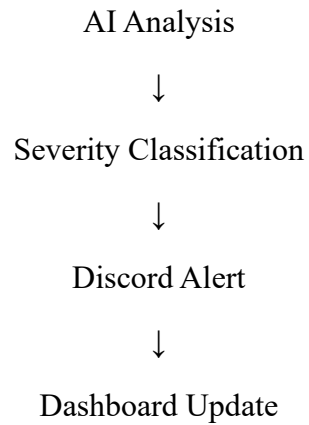


Queue Processing



Anomaly Detection





Error States

- Invalid Login Credentials
- Event Processing Failure
- AI Service Unavailable
- Database Connection Error
- Alert Delivery Failure

Redirect Logic

After Login → Dashboard

After Logout → Login

Unauthorized User → Login Page

4. BACKEND SCHEMA DESIGN

Database Overview

The backend uses SQLite for storing users, events, anomalies, and AI agent decisions.

Table: Users

Field	Type
id	Integer
username	String
password_hash	String
role	String

Field	Type
created_at	Timestamp

Table: Events

Field	Type
id	Integer
timestamp	Timestamp
source	String
value	Float
status	String

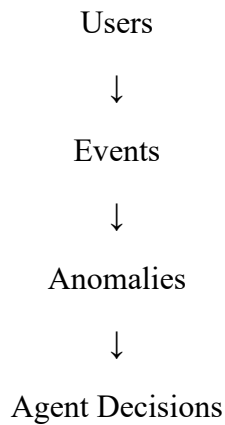
Table: Anomalies

Field	Type
id	Integer
event_id	Integer
z_score	Float
severity	String
explanation	Text

Table: Agent Decisions

Field	Type
id	Integer
anomaly_id	Integer
recommendation	Text
decision	String

Entity Relationships



Security Model

- JWT Authentication
- Password Hashing
- Role-Based Access Control
- Protected API Endpoints

5. UI/UX DESIGN BRIEF

Design Philosophy

Professional monitoring dashboard with a modern analytics-focused interface.

Dashboard Components

KPI Cards

- Total Events
- Total Anomalies
- Active Alerts
- Processing Rate

Analytics Charts

- Event Timeline
- Anomaly Trend
- Severity Distribution
- Alert Frequency

Tables

- Event Logs
- Anomaly Records
- Alert History

Responsive Design

- Desktop Support
- Tablet Support
- Mobile Support

Accessibility

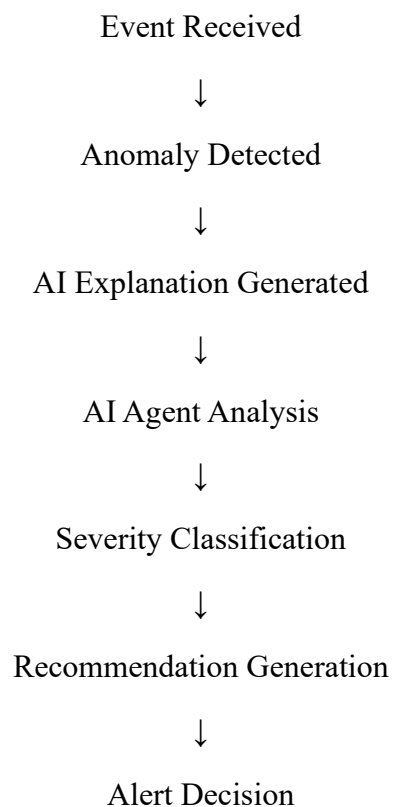
- Keyboard Navigation
- High Contrast Support
- Screen Reader Compatibility

6. AI NOTES & INTELLIGENCE DOCUMENTATION

AI Overview

The AI subsystem uses Llama 3.1 through Ollama to provide explanations, severity analysis, and actionable recommendations.

AI Workflow



Example 1 – Event Analysis

Event Value: 5000

Normal Range: 100 – 200

Result:

The event value significantly exceeds the normal operating range and is classified as anomalous.

Example 2 – Severity Classification

Z-Score: 4.5

Severity:

CRITICAL

Recommended Action:

Immediate investigation required.

Example 3 – Alert Decision

Condition:

Critical anomaly detected.

Action:

Trigger Discord Alert.

Severity Levels

LOW

Minor deviation from normal behavior.

MEDIUM

Moderate anomaly requiring observation.

HIGH

Significant anomaly requiring investigation.

CRITICAL

Severe anomaly requiring immediate action.

Benefits of AI Agent

- Faster incident response
- Reduced manual analysis

- Automated recommendations
- Improved operational visibility
- Better anomaly understanding

Conclusion

The AI-powered Streaming Event Anomaly Detector provides real-time monitoring, anomaly detection, intelligent explanations, and automated alerting. The platform enhances operational awareness and enables rapid response to abnormal system behavior through AI-driven decision support.